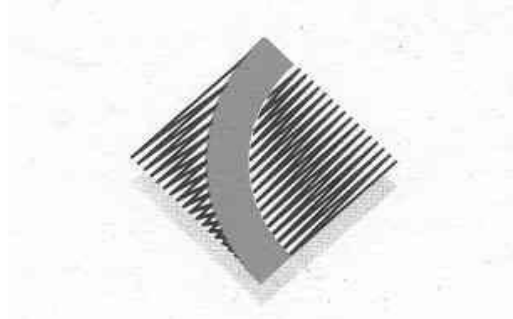


ML
MK
DH
AB
KSIN
KSAX
JI
CF
AM/
YR
BP

Name: _____
Class: 11MTA__ OR 11MTX__
Teacher: _____

CHERRYBROOK TECHNOLOGY HIGH SCHOOL



**2018
YEAR 11**

AP2 EXAMINATION MATHEMATICS

*Time allowed - 2 HOURS
(Plus 5 minutes reading time)*

DIRECTIONS TO CANDIDATES:

SECTION I: Use the multiple-choice answer sheet provided.

SECTION II:

- Each question is to be commenced in a new booklet clearly marked Question 11, Question 12, etc on the top of the page.
- All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- If you do not attempt a question, you must submit a blank booklet clearly indicating the question number, your name and class
- All questions should be placed in order - multiple-choice answer sheet followed by Question 11 to 14.
- NESA approved calculators may be used.
- The NESA Reference Sheet is provided.

Section I

10 marks

Attempt all questions

Allow about 15 minutes for this section

Use the multiple choice answer sheet for Questions 1 - 10

1. Which of the following is an expression for $x^2 - 5x - 6$?

(A) $(x - 2)(x + 3)$
(B) $(x + 1)(x - 6)$
(C) $(x - 1)(x + 6)$
(D) $(x + 2)(x - 3)$

2. Which of the following are all rational numbers?

(A) $-0.13, 2\sqrt{9}, \frac{22}{7}, 0$

(B) $3\pi, -0.13, \frac{22}{7}, 0$

(C) $\frac{22}{7}, \sqrt{5}, 2\sqrt{9}, 3\pi$

(D) $\sqrt{5}, 3\pi, -0.13, \frac{22}{7}$

3. Evaluate $\lim_{x \rightarrow 4} \frac{x^2 - 3x - 4}{x - 4}$

(A) 0
(B) 1
(C) 5
(D) Undefined

4. Solve $\tan \theta = \sqrt{3}$, $0 \leq \theta \leq 360^\circ$

(A) $60^\circ, 120^\circ$
(B) $60^\circ, 240^\circ$
(C) $30^\circ, 210^\circ$
(D) $30^\circ, 150^\circ$

5. Simplify $\sin^2 x - \frac{\tan^2 x + 1}{\sec^2 x}$
- (A) $-\cos^2 x$
(B) $\cos^2 x$
(C) $\sin^2 x$
(D) $\tan^2 x$
6. Consider $f(x) = \frac{6}{x}$ and $g(x) = 2x + 4$.
What are the values for x for which $f(x) = g(x)$?
- (A) $x = -1$ or $x = 3$
(B) $x = -1$ or $x = -3$
(C) $x = 1$ or $x = 3$
(D) $x = 1$ or $x = -3$
7. If $\log_x 3 = 2$ and $\log_x 5 = 3$, evaluate $\log_3 15$.
- (A) $\frac{5}{2}$
(B) 3
(C) 5
(D) 6
8. Which of the following is the equation of a circle with centre $(-1, 0)$ and radius 3?
- (A) $x^2 + 2x + y^2 = 2$
(B) $x^2 - 2x + y^2 = 8$
(C) $x^2 - 2x + y^2 = 2$
(D) $x^2 + 2x + y^2 = 8$

9. Differentiate $y = \sqrt[4]{2x^2 - 8}$

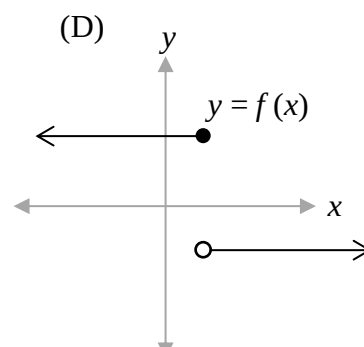
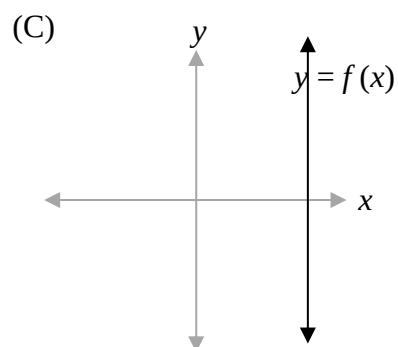
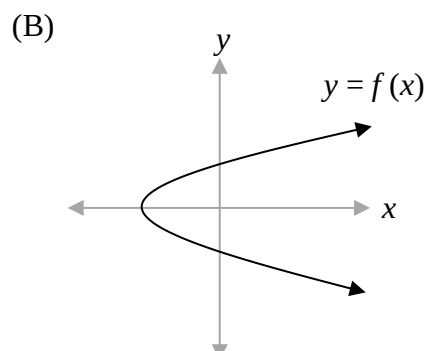
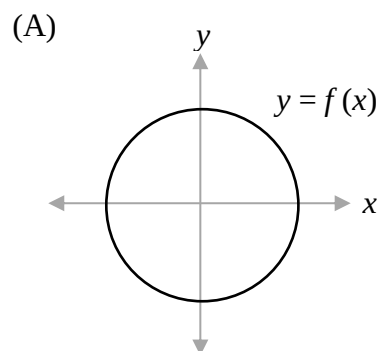
(A) $\frac{1}{4}\sqrt[4]{(2x^2 - 8)^3}$

(B) $\frac{x}{\sqrt[4]{(2x^2 - 8)^3}}$

(C) $x^4\sqrt{(2x^2 - 8)^3}$

(D) $\frac{1}{16x^4\sqrt{(2x^2 - 8)^3}}$

10. Which of the following graphs is a function?



END OF SECTION I

Section II

60 marks

Attempt Questions 11 – 14

Allow about 1 hours and 45 minutes for this section

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 Marks) Start a new writing booklet **Marks**

(a) Simplify $\sqrt{32} - \sqrt{18} + \sqrt{2}$ **1**

(b) Evaluate $|30 - 12 \times 3| - |-30 + 12 \times 3|$ **1**

(c) Solve $9^{3y+1} = 27$ **1**

(d) Simplify and leave your answer as a positive index. **2**

$$\left(x^{-1} + \frac{1}{y^{-2}}\right)^{-1}$$

(e) Evaluate $\frac{\sqrt[3]{2.75 \times 10^8}}{7.1 \times 5}$ correct to four significant figures. **1**

(f) Factorise $a^3 - 8b^3$ **1**

(g) Find the integers a and b such that $\frac{\sqrt{5}}{2+\sqrt{5}} = a + b\sqrt{5}$. **2**

(h) Solve $3x^2 - 10x + 3 = 0$ **2**

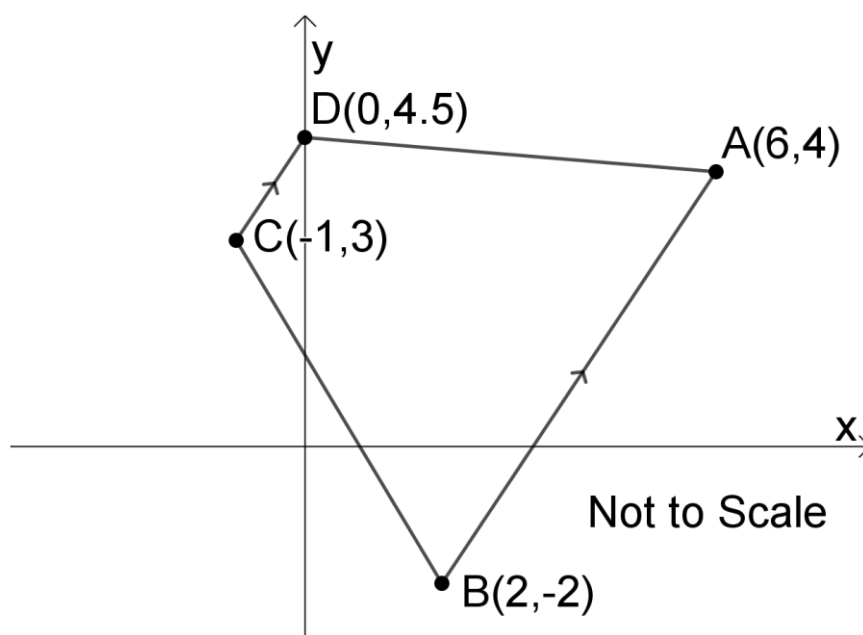
(i) Solve $|2x - 3| \geq 7$ **2**

(j) Express $\frac{x-1}{5} - \frac{2x-3}{9}$ as a fraction in its simplest form. **2**

End of Question 11

Question 12 (15 Marks)	Start a new writing booklet	Marks
-------------------------------	------------------------------------	--------------

- | | | |
|--|--|---|
| (a) Factorise completely $2x^2 - 2 - y + x^2y$ | | 2 |
| (b) Solve simultaneously $5x - 2y = 8$ and $2x - y = 5$ | | 2 |
| (c) Sketch the curve $y = 3\sin 2x$ for $0 \leq x \leq 360^\circ$. | | 2 |
| (d) The points $A(6, 4)$, $B(2, -2)$, $C(-1, 3)$ and $D(0, 4.5)$ are plotted on a number plane. AB is parallel to CD . | | |

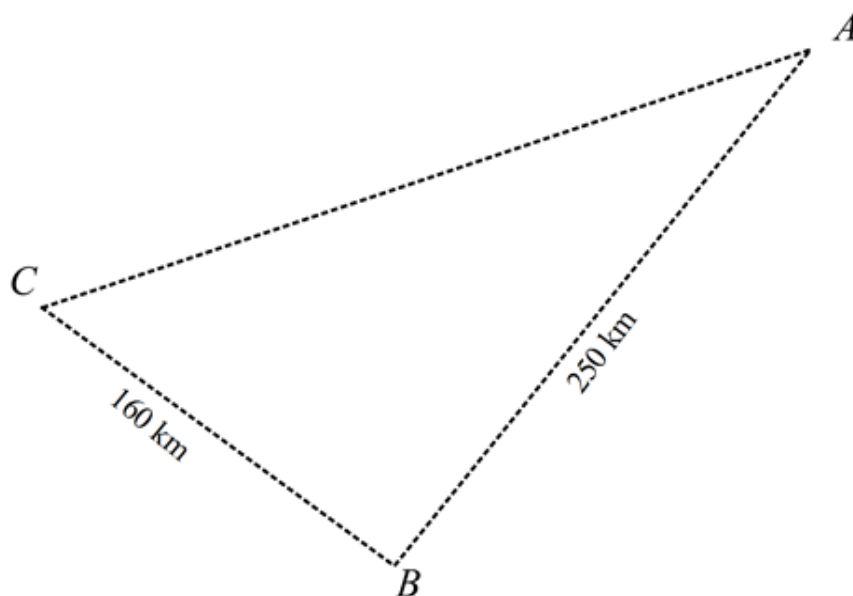


- | | | |
|---|--|---|
| (i) Find the exact length of AB . | | 1 |
| (ii) Find the gradient of AB . | | 1 |
| (iii) Show that the equation of AB is $3x - 2y - 10 = 0$. | | 1 |
| (iv) Calculate the perpendicular distance from C to AB , in exact form. | | 2 |
| (v) Hence, determine the area of triangle ABC . | | 1 |

Question 12 continues on the next page

- (e) Three towns, A , B and C are located as shown on the diagram below. From town C , the bearing of A is 072° and the bearing of B is 128° . Town B is 160 km from C and 250 km from A .

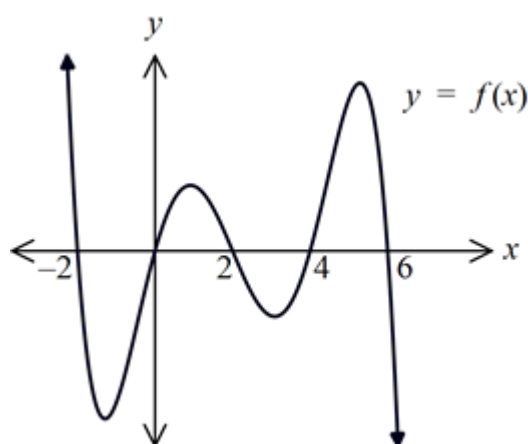
3



Calculate the bearing of B from A .

End of Question 12

- (a) Differentiate from first principles $f(x) = x^2 - 4x + 8$ **3**
- (b) Differentiate:
- (i) $y = (x^2 - 3)(x - 4)$ **2**
- (ii) $y = \sqrt{x} + \frac{1}{x^2}$ **2**
- (iii) $y = \frac{x}{7 - 3x}$ **2**
- (c) Write the equation of the normal to the curve $y = 6x^2 - x + 2$ at the point where $x = 1$ in general form. **3**
- (d) The tangent to the curve $y = x^3 - 3x^2$ at a point A has a gradient of -3 . Find the coordinates of the point A. **2**
- (e) Sketch the gradient function of the following graph. **1**



End of Question 13

Question 14 (15 Marks) Start a new writing booklet **Marks**

(a) State the domain and range of the function $f(x) = \sqrt{4 - x^2}$. **2**

(b) Determine algebraically if the function $f(x) = \frac{x^2}{x-1}$ is even, odd or neither **2**

(c) A function is defined by the rule:

$$f(x) = \begin{cases} x - 3, & x \leq -3 \\ 2x + 2, & -3 < x < 0 \\ x^2, & x \geq 0 \end{cases}$$

(i) Find $f(2) + f(-2) + f(-5)$ **2**

(ii) Sketch the above function. Show all intercepts. **3**

(d) Consider the function $f(x) = \frac{x}{2x^2 + b}$. **2**

Given that $f(-1) = f(1)$, find the value of b .

(e) Find the values of the constant p for which $\lim_{x \rightarrow p} \frac{x-p}{x^2-p^2} = 2p$ is correct. **2**

(f) Sketch the region where the graphs of $x + y + 1 > 0$ and $y \leq 1 - x^2$ hold simultaneously. **2**

End of Examination

Student Name.....

Class 11MTA..... or 11MTX.....

2018 CTHS MATHEMATICS AP2 EXAMINATION

Section I Answer Sheet

Attempt Questions 1-10

Allow about 15 minutes for this section.

Use the multiple choice answer sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think that you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

☒ ☒ ☐ ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

☒ ☒ ☐ ☐
correct

1. A ☐ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☐
5. A ☐ B ☐ C ☐ D ☐
6. A ☐ B ☐ C ☐ D ☐
7. A ☐ B ☐ C ☐ D ☐
8. A ☐ B ☐ C ☐ D ☐
9. A ☐ B ☐ C ☐ D ☐
10. A ☐ B ☐ C ☐ D ☐

MC	/10
Q11	/15
Q12	/15
Q13	/15
Q14	/15
Total	/70

AP2 Mathematics

Section 1

$$1. \quad x^2 - 5x - 6 \\ = (x-6)(x+1) \quad \boxed{B}$$

$$2. \quad \boxed{A}$$

$$3. \quad \lim_{x \rightarrow 4} \frac{x^2 - 3x - 4}{x - 4} \\ = \lim_{x \rightarrow 4} \frac{(x-4)(x+1)}{x-4}$$

$$= \lim_{x \rightarrow 4} (x+1)$$

$$= 4+1$$

$$= 5 \quad \boxed{C}$$

$$4. \quad \tan \theta = \sqrt{3}$$

$$\theta = 60$$

$$\therefore \theta = 60 \text{ or } 240 \quad \boxed{B}$$

$$5. \quad \frac{\sin^2 x - \tan^2 x + 1}{\sec^2 x}$$

$$= \frac{\sin^2 x - \sec^2 x}{\sec^2 x}$$

$$= \sin^2 x - 1$$

$$= -\cos^2 x \quad \boxed{A}$$

$$6. \quad \frac{b}{x} = 2x + 4$$

$$b = 2x^2 + 4x$$

$$2x^2 + 4x - 6 = 0$$

$$2(x^2 + 2x - 3) = 0$$

$$2(x+3)(x-1) = 0$$

$$\therefore x = -3 \text{ or } x = 1 \quad \boxed{D}$$

$$7. \quad \log_3 15 = \frac{\log_x 15}{\log_x 3}$$

$$= \frac{\log_x (5 \times 3)}{\log_x 3}$$

$$= \frac{\log_x 5 + \log_x 3}{\log_x 3}$$

$$= \frac{3 + 2}{2}$$

$$= \frac{5}{2} \quad \boxed{A}$$

$$8. \quad \boxed{D}$$

$$9. \quad y = (2x^2 - 8)^{\frac{1}{4}}$$

$$\frac{dy}{dx} = \frac{1}{4} (2x^2 - 8)^{-\frac{3}{4}} \times 4x$$

$$= x (2x^2 - 8)^{-\frac{3}{4}}$$

$$= x$$

$$= \sqrt[4]{(2x-8)^3} \quad \boxed{B}$$

$$10. \quad \boxed{D}$$

Section 2

Question 11.

a) $\sqrt{32} - \sqrt{18} + \sqrt{2}$

$$= 4\sqrt{2} - 3\sqrt{2} + \sqrt{2}$$

$$= 2\sqrt{2}$$

b) $|30 - 12 \times 3| - |-30 + 12 \times 3|$

$$= |30 - 36| - |-30 + 36|$$

$$= |-6| - |6|$$

$$= 6 - 6$$

$$= 0$$

c) $9^{3y+1} = 27$

$$(3^2)^{3y+1} = 3^3$$

$$6y+2=3$$

$$6y=1$$

$$y = \frac{1}{6}$$

d) $(x^{-1} + \frac{1}{y^2})^{-1}$

$$= (\frac{1}{x} + y^2)^{-1}$$

$$= (\frac{1+xy^2}{x})^{-1}$$

$$= \frac{x}{1+xy^2}$$

(e) $\sqrt[3]{2.75 \times 10^6}$

$$7.1 \times 5$$

$$= 18.318189...$$

$$\approx 18.32$$

f) $a^3 - 8b^3 = a^3 - (2b)^3$

$$= (a - 2b)(a^2 + 2ab + 4b^2)$$

g) $\frac{\sqrt{5}}{2+\sqrt{5}} \times \frac{2-\sqrt{5}}{2-\sqrt{5}}$

$$= \frac{2\sqrt{5} - 5}{4 - 5}$$

$$= -2\sqrt{5} + 5$$

$$\therefore \underline{a = 5}, \underline{b = -2}$$

h) $3x^2 - 10x + 3 = 0$

$$3x^2 - 9x - x + 3 = 0$$

$$3x(x-3) - (x-3) = 0$$

$$(3x-1)(x-3) = 0$$

$$\therefore x = \frac{1}{3} \text{ or } x = 3$$

$$i) |2x-3| \geq 7$$

must have or or comma.

$$2x-3 \geq 7$$

(or)

$$-(2x-3) \geq 7$$

$$2x \geq 10$$

$$-2x+3 \geq 7$$

$$x \geq 5$$

$$-2x \geq 4$$

$$x \leq -2$$

$$j) \frac{x-1}{5} - \frac{2x-3}{9}$$

$$= \frac{9(x-1) - 5(2x-3)}{45}$$

$$= \frac{9x-9-10x+15}{45}$$

$$= \frac{6-x}{45}$$

Question 12

a) $2x^2 - 2 - y + x^2y$

$$= 2(x^2 - 1) + y(x^2 - 1)$$

$$= (x^2 - 1)(y + 2)$$

$$= (x - 1)(x + 1)(y + 2)$$

Lots of students missed the +

b) $5x - 2y = 8$ — (1)

$$2x - y = 5$$
 — (2)

rearrange (2)

$$y = 2x - 5$$
 — (3)

Sub (3) into (1)

$$5x - 2(2x - 5) = 8$$

$$5x - 4x + 10 = 8$$

$$x = -2$$

Sub $x = -2$ into (2)

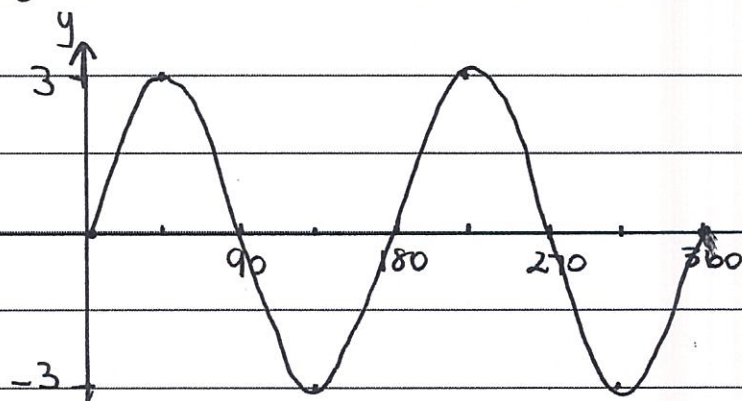
$$2(-2) - y = 5$$

$$-4 - y = 5$$

$$y = -9$$

$$\therefore x = -2, y = -9$$

c) $y = 3 \sin 2x$ $0 \leq x \leq 360$



This question was done poorly. Students need to review.

$$\begin{aligned}
 \text{d) (i) } d_{AB} &= \sqrt{(6-2)^2 + (4-(-2))^2} \\
 &= \sqrt{4^2 + 6^2} \\
 &= \sqrt{16+36} \\
 &= \sqrt{52} \\
 &= 2\sqrt{13}
 \end{aligned}$$

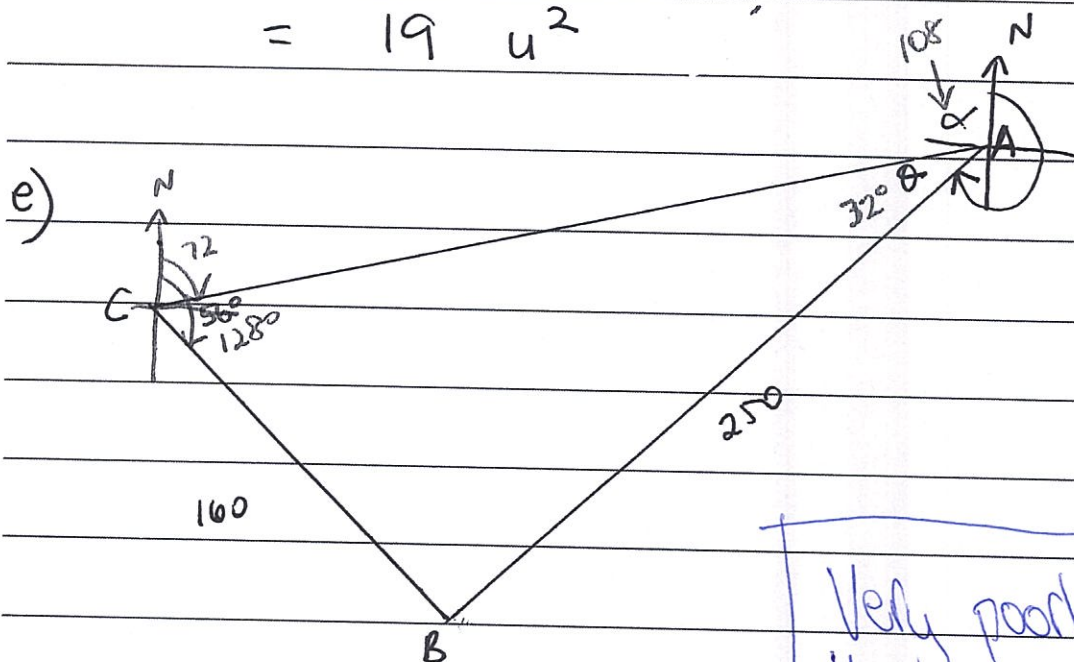
$$\begin{aligned}
 \text{(ii) } m_{AB} &= \frac{4-(-2)}{6-2} & \text{or} & & \frac{(-2)-4}{2-6} \\
 &= \frac{6}{4} & & & = \frac{-6}{-4} \\
 &= \frac{3}{2} & & & = \frac{3}{2}
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii) using pt } (6, 4), m &= \frac{3}{2} \\
 y - y_1 &= m(x - x_1) \\
 y - 4 &= \frac{3}{2}(x - 6) \\
 2(y - 4) &= 3(x - 6) \\
 2y - 8 &= 3x - 18 \\
 3x - 2y - 10 &= 0
 \end{aligned}$$

$$\begin{aligned}
 \text{(iv) } d &= \frac{|3x(-1) - 2y(3) - 10|}{\sqrt{3^2 + (-2)^2}} \\
 &= \frac{|-3 - 6 - 10|}{\sqrt{13}} \\
 &= \frac{|-19|}{\sqrt{13}} \\
 &= \frac{19}{\sqrt{13}}
 \end{aligned}$$

Had to show correct substitution of (-2) not 2.

$$\begin{aligned}
 (v) \quad A &= \frac{1}{2} bh \\
 &= \frac{1}{2} \times \frac{19}{\sqrt{13}} \times \sqrt{52} \\
 &= \frac{1}{2} \times \frac{19}{\sqrt{13}} \times 2\sqrt{13} \\
 &= 19 \text{ u}^2
 \end{aligned}$$



$$\angle ACB = 56^\circ$$

$$\text{let } \angle BAC = \theta$$

$$\frac{\sin \theta}{160} = \frac{\sin 56}{250}$$

$$\sin \theta = \frac{160 \sin 56}{250}$$

$$\therefore \theta = 32^\circ$$

Very poorly answered
Had to have some form
of reasoning. Didn't have
to be a complete reason,
as shown below.
However, in future
Exams full reasoning
will be expected.

$$\angle \alpha + 72 = 180 \text{ (co-interior } \angle \text{s)}$$

$$\angle \alpha = 108^\circ$$

$$\begin{aligned}
 \therefore \text{The bearing of B from A} &= 360 - 32 - 108 \\
 &= 220^\circ \text{ T}
 \end{aligned}$$

Question 13

a) $f(x) = x^2 - 4x + 8$

$$f(x+h) = (x+h)^2 - 4(x+h) + 8$$

$$= x^2 + 2xh + h^2 - 4x - 4h + 8$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 4x - 4h + 8 - (x^2 - 4x + 8)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 4x - 4h + 8 - x^2 + 4x - 8}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2xh + h^2 - 4h}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{h}(2x + h - 4)}{\cancel{h}}$$

$$= \lim_{h \rightarrow 0} 2x + h - 4$$

$$= 2x + 0 - 4$$

$$= 2x - 4$$

* for (b)

must show

$\frac{dy}{dx}$ or similar - only
penalised once

(b) (i) $y = (x^2 - 3)(x - 4)$

$$\frac{dy}{dx} = (x^2 - 3) \times 1 + (2x) \times (x - 4)$$

$$= x^2 - 3 + 2x^2 - 8x$$

$$= 3x^2 - 8x - 3$$

$$u = x^2 - 3$$

$$u' = 2x$$

$$v = x - 4$$

$$v' = 1$$

$$(ii) y = \sqrt{x} + \frac{1}{x^2}$$

$$= x^{\frac{1}{2}} + x^{-2}$$

$$\frac{dy}{dx} = \frac{1}{2} x^{-\frac{1}{2}} - 2x^{-3}$$

$$= \frac{1}{2\sqrt{x}} - \frac{2}{x^3}$$

mark not deducted
for not rewriting

with positive indices
• a lot of students left
solution with

$$\frac{1}{2\sqrt{x}} + \frac{2}{x^3}$$

deducted.

$$(iii) y = \frac{x}{7-3x}$$

$$\begin{cases} u = x \\ u' = 1 \\ v = 7-3x \\ v' = -3 \end{cases}$$

$$\frac{dy}{dx} = \frac{(1)(7-3x) - (x)(-3)}{(7-3x)^2}$$

$$= \frac{7-3x+3x}{(7-3x)^2}$$

$$= \frac{7}{(7-3x)^2}$$

$$(c) y = 6x^2 - x + 2$$

$$\frac{dy}{dx} = 12x - 1$$

$$\text{at } x = 1,$$

$$\frac{dy}{dx} = 12(1) - 1$$

$$= 11$$

$$\therefore m = 11$$

$$m_{\text{normal}} = \frac{-1}{11}$$

$$\begin{aligned} y &= 6(1)^2 - 1 + 2 \\ &= 6 - 1 + 2 \\ &= 7 \end{aligned}$$

$$\text{pt } (1, 7) \text{ and } m_n = \frac{-1}{11}$$

$$y - 7 = \frac{-1}{11} (x - 1)$$

$$11y - 77 = -x + 1$$

$$x + 11y - 78 = 0.$$

$$d) \quad y = x^3 - 3x^2$$

$$\frac{dy}{dx} = 3x^2 - 6x$$

$$m = -3$$

$$\therefore \frac{dy}{dx} = -3$$

$$-3 = 3x^2 - 6x$$

$$3x^2 - 6x + 3 = 0$$

$$3(x^2 - 2x + 1) = 0$$

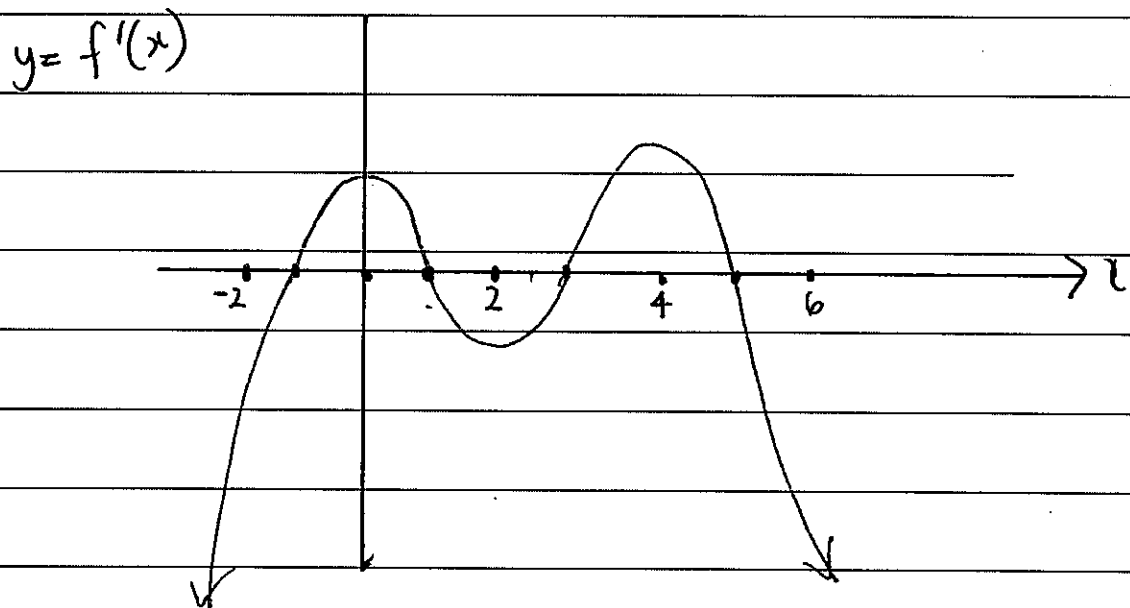
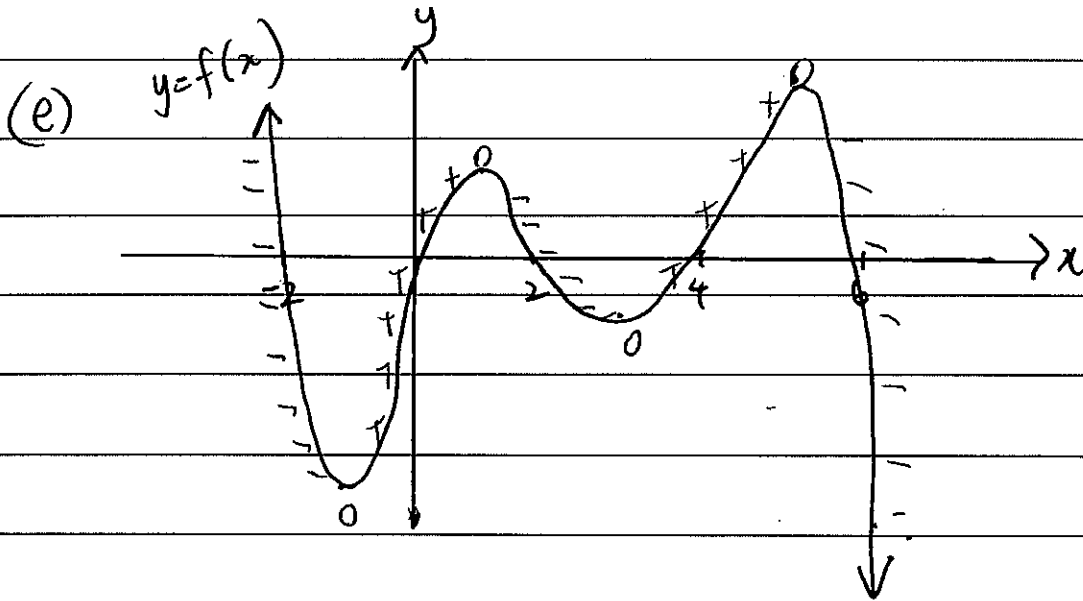
$$3(x-1)(x-1) = 0$$

$$\therefore (x-1)^2 = 0$$

$$x = 1$$

$$\begin{aligned} \text{when } x = 1, \quad y &= 1^3 - 3(1)^2 \\ &= 1 - 3 \end{aligned}$$

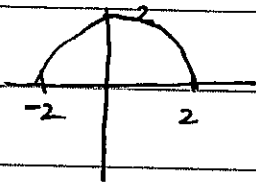
$$\text{point } A(1, -2) = -2$$



* This question was done poorly

Question 14.

(a) $f(x) = \sqrt{4-x^2}$



$\therefore$ Domain : $-2 \leq x \leq 2$

Range : $0 \leq y \leq 2$

(b) $f(x) = \frac{x^2}{x-1}$

$\therefore f(-x) = \frac{(-x)^2}{(-x)-1}$

$= \frac{x^2}{-x-1}$

$= -\frac{x^2}{x+1} \neq f(x)$
 $\neq -f(x)$

$\therefore$ neither odd nor even

needed to show in working

$-f(x) = -\left[\frac{x^2}{x-1}\right]$
 $= \frac{x^2}{1-x}$

(c) (i) $f(2) + f(-2) + f(-5)$

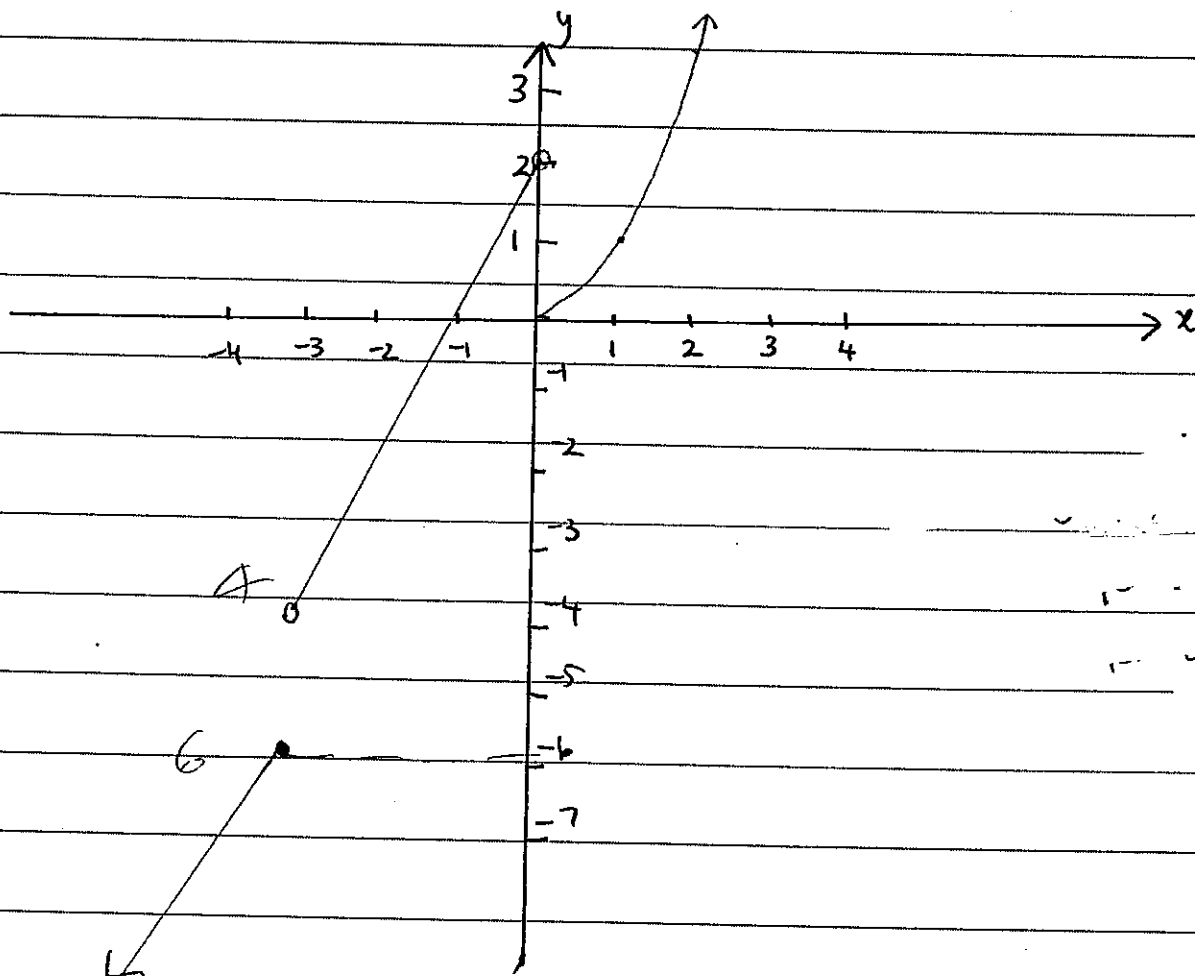
$= 2^2 + [2 \times (-2) + 2] + [(-5) - 3]$

$= 4 + (-2) + (-8)$

$= -6$

cf.m if only 1 error in this line

c(11)



(d) $f(x) = \frac{x}{2x^2 + b}$

$$f(-1) = \frac{-1}{2(-1)^2 + b}$$

$$= \frac{-1}{2+b}$$

$$f(1) = \frac{1}{2(1)^2 + b}$$

$$= \frac{1}{2+b}$$

As $f(-1) = f(1)$ then

$$\frac{-1}{2+b} = \frac{1}{2+b}$$

$$-(2+b) = 2+b$$

$$-2-b = 2+b$$

$$-4 = 2b$$

$$b = -2.$$

Q14

$$(c) \lim_{x \rightarrow p} \frac{x-p}{x^2-p^2} = 2p$$

$$= \lim_{x \rightarrow p} \frac{x-p}{(x-p)(x+p)}$$

$$= \lim_{x \rightarrow p} \frac{1}{x+p}$$

$$= \frac{1}{p+p}$$

$$= \frac{1}{2p}$$

$$\therefore \frac{1}{2p} = 2p$$

$$4p^2 = 1$$

$$p^2 = \frac{1}{4}$$

$$p = \pm \frac{1}{2}$$

Students received
no marks for
correct answer $p = \pm \frac{1}{2}$

when they found

$$\lim_{x \rightarrow p} \frac{1}{x+p} = 2p$$

then $\lim_{x \rightarrow p} 1 = 2p(x+p)$

$$(f) \quad x+y+1 > 0 \quad \text{and} \quad y \leq 1-x^2$$

